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Defining the Polycystin Pharmacophore Through HTS & Computational Biophysics

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Abstract

Background and Purpose: Polycystins (PKD2, PKD2L1) are voltage-gated and Ca²⁺-modulated members of the transient receptor potential (TRP) family of ion channels. Loss of PKD2L1 expression results in seizure-susceptibility and autism-like features in mice, whereas variants in PKD2 cause autosomal dominant polycystic kidney disease. Despite decades of evidence clearly linking their dysfunction to human disease and demonstrating their physiological importance in the brain and kidneys, the polycystin pharmacophore remains undefined. Contributing to this knowledge gap is their resistance to drug screening campaigns, which are hindered by these channels' unique subcellular trafficking to organelles such as the primary cilium. PKD2L1 is the only member of the polycystin family to form constitutively active ion channels on the plasma membrane when overexpressed.

Experimental Approach: HEK293 cells stably expressing PKD2L1 F514A were pharmacologically screened via high-throughput electrophysiology to identify potent polycystin channel modulators. In-silico docking analysis and mutagenesis were used to define the receptor sites of screen hits. Inhibition by membrane-impermeable QX-314 was used to evaluate PKD2L1's binding site accessibility.

Key Results: Screen results identify potent PKD2L1 inhibitors with divergent chemical core structures and highlight striking similarities between the molecular pharmacology of PKD2L1 and voltage-gated sodium channels. Docking analysis, channel mutagenesis, and electrophysiological recordings identify an open-state accessible lateral fenestration receptor within the pore, and a mechanism of inhibition that stabilizes the PKD2L1 inactivated state.

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Authors contributions using CRediT nomenclature and definitions.

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Data curation; Formal analysis; Methodology; Validation; Visualization, Writing – original draft.

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Conflict of Interest Statement:

The authors declare no conflict of interest

Conclusion and Implication: Outcomes establish the suitability of our approach to expand our chemical knowledge of polycystins and delineates novel receptor moieties for the development of channel-specific inhibitors in TRP channel research.

Keywords

autosomal polycystic kidney disease; ADPKD; PKD2L1; PKD2; polycystin; TRP channels; primary cilia; ion channels; molecular mechanisms; Pharmacology; Computational Biophysics; Molecular Docking; High-Throughput Electrophysiology

Regarding polycystin nomenclature:

The revised and current IUPHAR/BPS nomenclature creates ambiguity regarding the genetic identity of the polycystin family members of transient receptor potential ion channels (TRPP) when cross-referencing manuscripts that describe subunits using the former system.¹ Traditionally, the products of polycystin genes (e.g., PKD2) are referred to as polycystin proteins (e.g., polycystin-2). For simplicity and to prevent confusion, we will refer to the polycystin gene name rather than differentiating gene and protein with separate names—a nomenclature we have recently outlined.²

INTRODUCTION

Transient receptor potential (TRP) cation channels function as polymodal cellular sensors that open in response to a broad range of chemical, thermal and physical stimuli—resulting in electrical depolarization of nerves and other cell membranes (<https://www.guidetopharmacology.org/GRAC/FamilyDisplayForward?familyId=78>)³. Humans express a total of 28 different TRP channel proteins, which can be divided into seven subfamilies based on amino acid sequence homology: ankyrin, canonical, melastatin, mucolipin, polycystin and vanilloid⁴. Members of the polycystin class have six transmembrane segments which can assemble as homotetrameric channels that are voltage-gated and Ca²⁺-modulated^{5–10}. Revision of the IUPHAR/BPS nomenclature introduces ambiguity when identifying members of the polycystin family within the transient receptor potential ion channels (TRPP1–3), particularly when comparing studies that use the earlier naming system¹. To streamline communication and avoid confusion, we will consistently use the polycystins using their gene names (PKD2, PKD2L1, PKD2L2)—a convention we have recently proposed². Polycystin subunits also form heteromeric complexes with closely-related, eleven-transmembrane-segment proteins (PKD1, PKD1L1, PKD1L2), which contain both channel and adhesion g-protein-coupled-receptor components^{2,11,12}. Heteromeric polycystin channels are proposed to be cleavage or ligand activated, but their native functional properties are still being elucidated^{13,14}. Several members of the polycystin family show temporal expression during embryonic development (e.g. embryonic node) and localize to divergent organ tissues in adulthood, such as kidney principal cells and hippocampal neurons^{15,16}. Human clinical genetics and mouse transgenic studies indicate that polycystins are critically involved in disparate forms of cell physiology and function as “organizers” of cell differentiation during gastrulation^{17,18}. Human germline variants in renal polycystins (PKD1 and PKD2) are responsible for autosomal dominant polycystic

kidney disease (ADPKD), a monogenetic disorder affecting ≈ 13 million individuals, characterized by the proliferation of fluid-filled cysts that precipitate renal failure^{19–21}. In contrast, PKD2L1 variants are not reportedly associated with human disease, but loss of their expression causes hippocampal neuron hyperexcitability, which precipitates seizure susceptibility and autism spectrum disorder-like neurophenotypes in mice^{16,22,23}.

Despite their clear association with human disease and their roles in human development and cell physiology, the pharmacology of polycystins is largely undescribed. This is principally due to their unique subcellular localization, which makes them resistant to drug screening. The PKD2 renal polycystin forms functional channels in the primary cilia organelle membranes of principal cells, and when overexpressed in cell lines (<https://www.guidetopharmacology.org/GRAC/DatabaseSearchForward?searchString=TRPP1&searchCategories=all&species=none&type=all&comments=includeComments&order=rank>)^{6–8}. Importantly, they are electrically and chemically insulated from the cytosolic compartment, which prohibits their pharmacological screening using traditional, whole-cell patch clamp electrophysiology and soluble Ca^{2+} -indicator dyes^{14,24}. Like renal polycystins, PKD2L1 also functions as cilia channel, but it is primarily expressed in hippocampal interneurons and cerebrospinal fluid-contacting neurons in the spinal cord (<https://www.guidetopharmacology.org/GRAC/DatabaseSearchForward?searchString=TRPP2&searchCategories=all&species=none&type=all&comments=includeComments&order=rank>)^{16,25}. Unlike PKD2, the subcellular distribution of PKD2L1 is less restricted—forming channels in the neuronal soma or plasma membrane when transgenically expressed in cell lines¹⁴. In this report, we leverage this feature and use PKD2L1 as an archetypal polycystin channel to carry out high-throughput screen (HTS) electrophysiology for high-potency chemical matter against these elusive targets. Recent advancements in automated patch-clamp technology have made electrophysiology screening for TRP channel ligands a more viable approach²⁶. These systems provide $G\Omega$ -seal quality data, comparable to manual electrophysiology, but simultaneously voltage clamp hundreds of cells in planar configuration. Combined with automated liquid handling capabilities, these instruments allow for screening of entire chemical libraries and thorough assessments of a test article's potency and biophysical mechanism of action against PKD2L1. Our screen results identify four potent PKD2L1 inhibitors with divergent chemical core structures from separate drug classes, including local anesthetics and class 1C antiarrhythmics that target voltage-gated sodium channels (Na_v). In silico analysis of drug docking to PKD2L1 cryo-electron microscopy (cryo-EM) structures pinpoints high-affinity occupancy sites within the vestibule of the pore domain, which are conserved in Na_v channels. Our study spotlights the PKD2L1 pore vestibule and lateral fenestrations as promiscuous and state-dependent receptors for chemical antagonism of polycystin channels. Furthermore, it delineates a shared molecular pharmacology between Na_v and polycystins in mechanistic and structural terms, which are driven by state-dependent conformational changes, initiated by membrane depolarization.

METHODS

Cells, Vector Expression and Mutagenesis

18–24 hours prior to manual recordings, polycystin channel cDNA was co-transfected alongside enhanced green fluorescence protein (EGFP) cDNA using Lipofectamine 2000 (Invitrogen). To generate stable cells for automated recordings, HEK 293 cells (ATCC, CRL-1573, RRID:CVCL_0045) were infected with lentivirus expressing PKD2L1 F514A and IRES GFP under a CMV promoter (Vector Builder). After 48 hours, the resulting stable cells were selected with puromycin (Thermo Fisher Scientific) and maintained in Dulbecco's modified Eagle's medium (GIBCO) supplemented with 10% fetal bovine serum (Sigma-Aldrich) and 50 mg/mL streptomycin (Cytiva). Plasmids for PKD2L1 mutations were generated via site-directed mutagenesis of a WT PKD2L1 plasmid using the Q5 Site-Directed Mutagenesis Kit (New England Biolabs). Mutagenic primers were designed through the Q5 designer tool (New England Biolabs) and are listed in Table S5. For all engineered amino acid substitutions, successful mutagenesis was verified with whole-plasmid sequencing (Plasmidsaurus).

Chemical library.

High-throughput electrophysiology screen was performed using the 70 compounds included in the Screen-Well Ion Channel Ligand Library (BML-2805–0100, Enzo) and 30 individually sourced compounds. All compounds in the Enzo library had stock concentrations of 10mM in either DMSO or water, while individually sourced compounds were reconstituted to either 1 mM or 10 mM concentrations in DMSO or water (Table S1). Aliquots of each stock solution were prepared prior to screen experiments to avoid contamination and freeze-thaw degradation of original stocks. When not in use, stock solutions were stored at –80 °C and aliquots were stored at –20 °C. Prior to each run of HTS, test article working solutions were prepared fresh in assay buffer (extracellular recording solution) to achieve a final concentration of 1 μ M in recording wells. For concentration-response experiments, test articles were first diluted at log/half-log increments in DMSO/water to produce a series of five test concentrations. To ensure validity of screen-identified hits, subsequent concentration-response experiments were performed with externally-sourced samples. All compounds and sources can be found in Table S1.

Manual Patch Clamp

Whole-cell patch clamp recordings were performed with an AxioPatch 200B amplifier, Digidata 1550B digitizer, and pClamp 11 recording software (all Molecular Devices). Borosilicate glass capillaries (1B150F-4, World Precision Instruments) were fabricated using a P-1000 pipette puller (Sutter Instruments) to make recording electrodes with 2–5 M Ω tip resistances. Extracellular recording solution contained: 150 mM NaCl, 10 mM HEPES, and 2 mM CaCl₂. For experiments on WT PKD2L1 and F549 mutants, pipette solution contained: 80 mM CsMES, 20 mM NaCl, 15 mM BAPTA, 10 mM HEPES, 5 mM EGTA, and 2 mM MgCl₂. For experiments on PKD2L1 F514A, pipette solution contained: 110 mM CsF, 10 mM CsCl, 10 mM CsCl, 10mM HEPES, and 20mM EGTA. All solutions were adjusted to a final pH of 7.4 and an osmolarity of 300 mOsm ($\pm$ 5 mOsm) with mannitol. All experiments were performed at 21 °C. Tail currents were

elicited through a voltage-step protocol with a holding potential of -60 mV, a 100ms step to $+100$ mV, and a repolarization to -60 mV for wildtype PKD2L1. For PKD2L1 F514A, this was adjusted to a -80 mV holding potential and 120mV step to account for the altered voltage-dependence of opening, as previously reported³³. To minimize voltage error, cells were excluded from analyses if I_{leak} exceeded -200 pA at V_{holding} , or if R_{series} exceeded 10 M Ω . Normalized current inhibition was calculated from the ratio of pre-treatment current amplitudes (baseline) and post-treatment steady-state current amplitudes, expressed as I/I_{baseline} . For current-over-time data, tail current amplitudes were normalized by cell capacitance ($I_{\text{tail}}/C_{\text{cell}}$) to minimize variance from cell-to-cell differences in size and resulting current amplitudes. Concentration-response plots were fit to the Hill equation ($Y = \text{Base} + (\text{Max}-\text{Base})/(1+(\text{IC}_{50}/x)^{\text{HillSlope}})$) to evaluate the potency (IC_{50}) of inhibitory compounds. Voltage-dependence of inactivation was analyzed through fitting a two-state Boltzmann ($I/I_{\text{Max}} = 1/(1+\exp((V_{1/2}-V)/k))$) distribution to normalized tail current data generated by a pre-pulse protocol consisting of a 2-second variable prepulse (-60 to 100 mV), followed by a test pulse (100 mV). A gravity-fed extracellular bath perfusion system with flow rate $5-10$ mL/min was used to apply test compounds in manual patch configuration. Washout was omitted from analysis, as inconsistent desensitization after prolonged stimulus renders wash off highly unreliable. Further, drug-induced facilitation of the desensitized state (see Figure S2A–B dibucaine), further limits the accuracy of wash-off results.

Automated Patch Clamp

Whole-cell planar patch recordings were performed on a Syncropatch 384I (Nanion Technologies) automated patch clamp platform using S-type 384 well 1X chips with $2-4$ M Ω resistances (Nanion Technologies). Pulse protocols and data collection were controlled via PatchController384 v3.2.0.27 (Nanion Technologies). Extracellular recording solution contained: 140 mM NaCl, 10 mM HEPES, 5 mM glucose, 4 mM KCl, 2 mM CaCl_2 , and 1 mM MgCl_2 with a final pH of 7.4 , adjusted with NaOH. Seal enhancer solution contained: 80 mM NaCl, 35 mM CaCl_2 , 10 mM MgCl_2 , 10 mM HEPES, and 3 mM KCl with a final pH of 7.4 , adjusted with NaOH. Intracellular solution contained: 110 mM CsF, 10 mM CsCl, 10 mM CsCl, 10 mM HEPES, and 20 mM EGTA, with a final pH of 7.2 , adjusted with CsOH. All experiments were performed at 21 °C. Rigorous criteria were used to filter out poor-quality recordings prior to final data analysis. Cells qualified for final analysis if they met the following QC standards: Seal resistance ≥ 200 M Ω . Series resistance ≤ 20 M Ω . Baseline current between -100 and 100 pA. Tail peak amplitude between -200 pA and -10 nA. During standard planar patch experiments, 384 well chips were partitioned into 12 two-column sections, 10 of which contained test compounds, 1 of which contained a known blocker (LaCl), and one of which served as a DMSO control, for a total of 32 potential replicates per condition. Compounds were identified as hits if 1 μM treatment resulted in 1) a change from baseline to response amplitude of greater than 25% and 2) a statistically significant ($p < 0.0005$) difference between the mean response of treated and control cells, determined via unpaired t-test.

3D Modeling and In-silico Docking

All in-silico docking models were generated with AutoDock4 v4.2.6⁴². PKD2L1 open (PDB: 6DU8) and closed (PDB: 5Z1W) structures were prepared for docking analysis through removal of water molecules, the addition of polar hydrogens, and the addition of Kollman charges. Atoms were then assigned to an AutoDock 4 type. Ligand molecules were optimized in Avogadro, and all rotatable bonds were unlocked prior to docking. All molecules first underwent blind docking, during which the bounding box encompassed the entirety of the channel structure. Based on the results of blind docking, the bounding box in subsequent docking analyses was restricted to regions with the most stable docking conformations. Final docking results were obtained with 100 runs of the Lamarckian Genetic Algorithm under the following parameters: Population Size = 300, Number of Generations = 27000, and Number of Evaluations = 2.5–50 million depending on the torsion # for a given ligand.

Data Analysis.

Data were analyzed via a combination of DataControl384 (Nanion Technologies), pClamp 11 (Molecular Devices), Prism 8 (Graphpad), and Rstudio (Posit). Access to R-scripts used to process data is available upon request.

Design and Statistical Analysis.

Studies were designed to generate group of equal size, and were randomized by inclusion of cells from multiple passages and transfections. Blinding was unfeasible during experiments, as a single author was responsible for drug formulation, application, and recording of all responses. Where groups of unequal size are present, this is the result of exclusion of individual cells during quality control and data analysis (e.g. high leak, low voltage-resistance, CDM/CDD). Samples sizes for each dataset were based on previous experiments and author experience³¹. Statistical analyses were undertaken only for studies where each group size was at least $n = 5$. In one experiment (Figure S1C), two groups of $n = 4$ were included but not subjected to statistical analysis, as they represent previously described phenomena and are to be considered exploratory. Declared group sizes represent the number of independent values and statistical analyses were performed using these independent values (not technical replicates). Statistical methods used to determine significance are described in corresponding figure legends and reported as P values. Briefly, HTS datasets were analyzed using unpaired t-tests, with statistical significance (*) being stated if $P < 0.0005$. Additionally, effect size was calculated via Cohen's d and reported alongside all P values in Table S6. Outliers were included in data analysis and presentation for across all datasets except in Figure 1D (Lido 0.1 μM and 100 μM ; Gd^{3+} 10 μM and 100 μM), which were excluded from analysis and presentation after identification via Grubb's test ($\alpha = 0.05$). The number of outliers excluded are represented in Table S7 as numbers in red color text.

Materials availability statement.

All cell expression constructs used in this study are available without restriction upon written request to the corresponding author.

Data availability statement.

Data reported in this paper is deposited and will be made available after publication, without restriction through the NU library ARCH (<https://doi.org/10.21985/n2-f3zk-rq43>).

Declaration of Transparency and Scientific Rigor.

This Declaration acknowledges that this paper adheres to the principles for transparent reporting and scientific rigor of preclinical research as stated in the BJP guidelines for Design & Analysis as recommended by funding agencies, publishers and other organizations engaged with supporting research.

Nomenclature of Targets and Ligands

Key protein targets and ligands in this article are hyperlinked to corresponding entries in <http://www.guidetopharmacology.org>, and are permanently archived in the Concise Guide to PHARMACOLOGY 2021/22 (Alexander et al., 2021).

RESULTS AND DISCUSSION

HTS identifies potent PKD2L1 inhibitors with diverse chemical core structures.

Functional screens of TRP channels are frequently mired by spontaneous sensitization and desensitization phenomena—biological features that can lead to false-positive and false-negative interpretations of drug efficacy^{27–29}. Polycystins (PKD2 and PKD2L1) display both calcium-dependent modulation (CDM) and irreversible calcium-dependent desensitization (CDD) when natively expressed in cilia membranes or when overexpressed^{30,31}. High-throughput planar electrophysiology systems require low-capacity calcium buffering and the use of fluoride within their intracellular saline to support high-resistance seals in whole-cell configurations³². When testing HTS saline compositions under manual patch-clamp, transiently expressed PKD2L1 channels recorded from the plasma membrane of HEK cells exhibit rapid CDM and CDD phenomena. These manifest as a transient potentiation of tail currents activated by depolarization trains (–60mV to +100mV, 0.2Hz) and a subsequent loss (within five minutes) of all tail currents (Figure S1A, C). Previously, a complete alanine scan of the PKD2L1 pore domain identified the pore turret F514A mutant, which delays the onset of CDM and attenuates CDD, producing stabilized tail current magnitudes for over 15 minutes under HTS conditions (Figure S1B, C)³³. Before moving forward with the F514A mutant in our drug screening campaign, we validated its pharmacological sensitivity to previously identified PKD2L1 inhibitors (dibucaine, lidocaine, Gd³⁺, and La³⁺)³¹. The resulting concentration-response relationships show that against all four compounds evaluated, WT PKD2L1 and F514A channels display similar potency (1–2x) of current inhibition (IC₅₀) (Figure S2A–F, Table S2). Further, we verified the sensitivity of F514A to these same compounds under automated patch clamp (Nanion Technologies) and manual patch clamp (Molecular Devices) configurations and observed that for all inhibitors (except Gd³⁺) potencies against F514A remained consistent (Figure S3A–D, Table S2). We suspect the reduction of Gd³⁺ potency (IC₅₀ = 1 μM to 7 μM) was caused by precipitation of GdF₃ formed by electrolysis with the fluoride present in the HTS internal saline, and thus did not include additional trivalent ion species in our screening library. Notably, Hill slopes for

local anesthetics concentration-response curves in the automated configuration are shallower than those of manual patch configuration, perhaps resulting from current “rundown” during extended perfusion in the manual configuration, relative to rapid dosing in the automated configuration (Figure S3A–B, Table S2). Together, these results confirm the suitability of our approach— using a mutant channel F514A for electrophysiology-based HTS to identify pharmacological chemical modulators of polycystin channels.

Previously identified polycystin modulators exhibit low potency ($IC_{50} > 1 \mu M$)³¹. As such, we designed our screen to identify compounds with potent effects in the micromolar range in order to spotlight likely targets for prototype-based drug discovery. To assemble our polycystin screening library (100 compounds total), we combined small molecules from commercially available channel-ligand screening collections (SCREEN-WELL®, Enzo Life Sciences) with individually sourced prototypic or clinically used drugs (Millipore Sigma) with established channel pharmacology (Table S1). To identify ligands with reproducible and high potency efficacy against PKD2L1, we assessed the effects of each test compound at 1 μM concentration against 6–38 separate cells over several cell line passages. From this first-pass trial, we observed an average of 118 successful wells per chip (~31%) after QC filtering, and were able to identify a chemically divergent set of compounds (U-50488, propafenone, tetrandrine, and AM-92016) which significantly inhibited polycystin tail currents (Figure 1A–C, Table S6). Notably, U-50488³⁴, propafenone³⁵, and tetrandrine³⁶ have all been reported to show cross-reactivity with ion channels other than their canonical targets, suggesting these compounds may be particularly cross-reactive. Additional concentration-response experiments defined their potency of inhibition (IC_{50}) ranging from 1.0–3.8 μM (Figure 1D, Table S2). Notably none of the screened compounds enhanced PKD2L1’s voltage-activated tail currents and less than 4% had inhibitory properties, suggesting that as a drug target, the channel is not prone to indiscriminate chemical activation or inhibition. Among the identified PKD2L1 inhibitors, dibucaine, propafenone and tetrandrine are canonical voltage-gated sodium channel (Na_v) and calcium channel (Ca_v) inhibitors with antiarrhythmic and local anesthetic properties and established receptor sites within their molecular targets^{37–41}.

In silico structural analysis predicts receptor sites within the pore vestibule.

Given the chemical diversity of our identified potential ligands and the paucity of structural information on potential receptor sites within the PKD2L1 channel, we used previously determined cryo-EM structures to perform in-silico docking analyses for all four hits using AutoDock4.2⁴². Blind docking results exhibited a propensity toward stable binding within the PKD2L1 pore domain (data not shown). With this in mind, we performed targeted docking of the pore domain and generated clustering histograms for the resulting conformations (Figure 2A). U-50488 and tetrandrine docking produced conformational clusters with a narrow range of predicted binding energies and averages of -7.16 kcal/mol and -8.65 kcal/mol, respectively (Figure 2A–B). In contrast, propafenone and AM-92016 analyses generated docking conformations with greater variance in binding energies, and averages of -4.65 kcal/mol and -6.65 kcal/mol, respectively (Figure 2A–B). Next, we visualized the 10 highest ranked conformations for each compound to better interpret their proposed binding sites. These models predict that for U-50488 and propafenone, stable

binding conformations cluster within the upper surface of the pore vestibule (Figure S4A–B). For tetrandrine, the highest ranked conformations form a cluster within the bottom of the pore vestibule (Figure S4C). AM-92016 shows the least consistent grouping, with predicted binding conformations distributed throughout the pore vestibule (Figure S4D). To gauge the relevance of these docking conformations to our in vitro data, we compared AutoDock-predicted K_i values to our in vitro IC_{50} values. Predicted K_i 's for the most stable conformations of U-50488 ($K_i = 0.9 \mu\text{M}$), tetrandrine ($K_i = 0.3 \mu\text{M}$), and AM-92016 ($K_i = 1.5 \mu\text{M}$) either fell in agreement with or predicted a greater potency than measured in vitro ($IC_{50} = 1 \mu\text{M}$, $2.1 \mu\text{M}$, and $3.8 \mu\text{M}$, respectively) (Figure 1B, Table S2, Table S3). In contrast, the predicted K_i of propafenone ($K_i = 11.9 \mu\text{M}$) showed a 10-fold lower potency than observed in vitro ($IC_{50} = 1 \mu\text{M}$) (Figure 1B, Table S2, Table S3). These results are consistent with the known limitations of the software when docking molecules with high torsion numbers, and reflect the loss in predictive power that accompanies complex molecule docking. Although a useful tool in highlighting potential targets, docking analyses are limited by a variety of factors, including the assumption of a rigid receptor and limited accounting for water/solvent effects. Despite this, these results suggest that for three of our four compounds, the predicted dynamics of their most stable docking conformations show consistency with our in vitro observations. Notably, for all compounds except tetrandrine, the top-ranked docking conformations localized to the upper pore domain, implying the existence of a binding site with favorable interactions for multiple drug classes (Figure 2B). Utilizing these top-ranked conformations as a guide, we analyzed our docked structures with the Protein-Ligand Interaction Profiler (PLIP)⁴³. Across all four drug-bound models, hydrophobic interactions provided the bulk of stabilizing forces (Figure 2C). Pore vestibule residues F556 and F549 participate in pi-stacking interactions with tetrandrine's aromatic rings, while the backbone oxygen molecules of I520 and L552 form hydrogen bonds with propafenone and AM-92016. Interestingly, the aromatic rings of top-ranked conformations for U-50488, propafenone, and AM-92016 are coordinated by sidechain phenyls of F549 and F524. The proximity to these residues suggests the potential to swap to alternate conformations with the ability to pi-stack. Together, the modeling analyses suggest the pore vestibule is a shared receptor site for a diverse set of chemical inhibitors of PKD2L1 channels.

PKD2L1 lateral fenestration residues forms a local anesthetic-antiarrhythmic receptor site.

As demonstrated from our pharmacological screen, PKD2L1 currents are sensitive to sodium channel inhibitors (dibucaine, lidocaine, propafenone)—drugs that are clinically used as local anesthetics (LAs) and/or class 1C antiarrhythmics (AAs). The molecular location of their binding site was first described in mutagenesis studies, but subsequent structural determinations in prokaryotic Na_v orthologs (e.g. Na_vAb) have defined the upper pore vestibule as the receptor site responsible for LA-AA drug action⁴⁴. Here, these molecules interact with residues lining the lateral fenestrations within the Na_v pore domain, formed by S5-S6 transmembrane helical segments⁴⁵. The pore domains of homotetrameric PKD2L1 and Na_v channels are structurally homologous (32% by residue identity and 58% by character), and both channel types have lateral fenestrations with comparable dimensions in their high-resolution structures (Figure 3A)⁴⁶. Within the lateral fenestrations, several key Na_v LA-AA receptor site residues are conserved in PKD2L1 channels and retain

their vestibule-facing structural orientation, as well as the general hydrophobicity of their neighboring residues (Figure 3B–C). To predict if a Na_v-like dibucaine binding site is indeed shared by PKD2L1, we performed in silico docking analysis and observed enhanced affinity for the closed state channel structure ($K_i = 5.9 \mu\text{M}$) compared to open state ($K_i = 10.5 \mu\text{M}$) at overlapping receptors sites within the upper pore vestibule, near the lateral fenestrations (Figure 4A, B). Similar to results from our screen hits, PLIP analysis predicted that binding of dibucaine within PKD2L1 relies primarily on hydrophobic interactions between the ligand and upper pore residues (Figure 4B). Additionally, residue F549, analogous to known drug binding Na_v1.5 residue F1760⁴⁷, is predicted to participate in both hydrogen bonding and pi-stacking with dibucaine (Figure 4B). Considering these docking results, we asked whether mutagenesis of this site in WT PKD2L1 could alter drug antagonism in functioning channels. A previous mutagenic screening of the PKD2L1 pore domain reported that F549A, V, and S mutations all result in the complete loss of channel function— therefore, these could not be tested in our electrophysiology assays³³. Guided by this knowledge, we chose to target less disruptive substitutions in the WT channel, including hydrophobic residues with comparable sidechain lengths (F549I, L) and aromatic residues that maintained the potential for pi-stacking (F549H, W). Of these, the I, L, and H substitutions all produced functional currents when expressed in mammalian cells, but showed altered rates of closure, relative to WT PKD2L1 (Figure 4C). Closure rates were fastest in F549L ($\tau_{\text{close}} = 5.7 \text{ ms}$) and slowest in WT PKD2L1 ($\tau_{\text{close}} = 33.9 \text{ ms}$). When comparing the dibucaine potency of current inhibition, F549L ($\text{IC}_{50} = 10.5 \mu\text{M}$) produced little to no change compared to wildtype ($\text{IC}_{50} = 8.3 \mu\text{M}$), whereas H ($\text{IC}_{50} = 17.2 \mu\text{M}$) and I ($\text{IC}_{50} = 76.2 \mu\text{M}$) substitutions showed a 2 to 10-fold reduction in potency (Figure 4D). Using the change in IC_{50} as an apparent change in K_i , we calculated a loss of binding affinity (ΔG) of 5.3 kcal/mol for the isoleucine mutation (Figure 4E, Table S4). In combination with the observed closure rates for each channel variant, these results indicate that the loss of dibucaine affinity in the I substitution likely does not result from changes in channel closure or access, but are the direct results of a loss in binding affinity. Due to the overlapping docking conformations for U-50488 and dibucaine, we evaluated U-50488 inhibition of F549I and observed a 2x loss in potency, relative to wildtype PKD2L1 (Figure S6A–B, Table S2). Together, these results support the conclusion that PKD2L1 and Na_vs share conserved lateral fenestrations in the pore domains, which form receptor sites for local anesthetics, antiarrhythmics and other chemical inhibitors, such as U-50488.

State-dependent mechanism of PKD2L1 channel inhibition.

Na_v antagonism by LAs and AAs is characterized by frequency-dependent (i.e., use-dependent) inhibition, a feature that underlies their clinical efficacy and clinical safety^{44,48}. Na_vs under high-frequency depolarizations show enhanced LA-AA receptor potency and accelerated kinetics of inhibition, which suggests greater ligand access to the lateral fenestration receptor site in the open or inactivated states. These characteristics allow for selective inhibition of aberrantly hyperexcitable neurons or cardiomyocytes, without impairing the rising phase of action potentials from unaffected cells⁴⁸. To determine if local anesthetic access to the PKD2L1 lateral fenestration receptor is also state-dependent, we applied 100 ms depolarizations (100 mV) at different frequencies in presence of 10 μM dibucaine (Figure 5A). The rate of inhibition was three-fold faster at 2 Hz ($\tau = 7.4 \text{ s}$)

than at 0.2 Hz ($\tau = 24.7$ s) stimulus, consistent with frequency-dependent inhibition and greater drug accessibility to the lateral fenestration receptor in the open state. Examination of the cryo-EM structures of WT PKD2L1 and modeled F549 mutants indicate that the lateral fenestration becomes sealed in the open state, suggesting small molecules become trapped when the channel transitions out of the closed state (Figure S5A–B). Once bound to the lateral fenestration receptor, we hypothesized two molecular mechanisms which could result in channel inhibition. LAs (or other analogues) could either bind and directly plug the ion conducting pathway of the channel or bind and stabilize PKD2L1's non-conducting inactivated state. When examining our models of dibucaine bound to the fenestration receptor (open state), orientation of the drug molecules did not encroach into central vestibule, suggesting that the mechanism is not blockade of the incoming cations. To assess the alternative hypothesis, we compared PKD2L1 voltage-dependent inactivation in control and dibucaine-treated conditions using a 2 second variable prepulse (–60 to 100 mV), followed by a test pulse (100 mV) to open the available channels (Figure 5B–C). Here, we observed a significant hyperpolarizing shift (–69 mV) in voltage-dependent inactivation after dibucaine treatment (Inactivation $V_{1/2}$, control = 49 mV, dibucaine = –20 mV). Taken together, these results demonstrate LA access the PKD2L1 lateral fenestration receptor site in the open-inactivated states. Once they occupy the lateral fenestration receptor site, LAs stabilize the non-conducting inactivated state of the channel, which causes inhibition of the current (Figure 6A).

Dibucaine and the other screen-identified inhibitors have high octanol-water partition coefficients ($\log P > 3$) and can permeate cell membranes. Since the lateral fenestration is located within the transmembrane vestibule, we propose that drug molecules access this receptor site either by traversing within the cell membrane and through the lateral fenestration (hydrophobic pathway) or by passing through membrane to access the open pore from the cytosolic side (hydrophilic pathway) (Figure 6B). To test drug access to the PKD2L1 lateral fenestration receptor, we explored PKD2L1 current inhibition by membrane impermeable QX-314 (a lidocaine derivative), previously used to define the mode of Na_v inhibited by local anesthetics. When introduced to the internal side of channel via the patch electrode, QX-314 produced an immediate and total inhibition of PKD2L1 currents, reminiscent of results from classical experiments characterizing Na_v channel inhibition (Figure S7A)⁴⁴. To determine if QX-314 could access the lateral fenestration receptor by passing through the inner gate, we measured the pore radii of PKD2L1 closed and open structures using the HOLE program (Figure S7B, C)⁴⁹. Here, the inner gate radius is restricted to 0.55 Å in the closed state and dilates to 2.6 Å upon channel opening, which is wide enough to pass molecules from the cytosolic side and may explain the observed open state-dependence of local anesthetic inhibition. When applied to the external side of the channel via the bath solution, QX-314's effect on PKD2L1 currents were limited, requiring > 7 minutes to reach steady state inhibition (Figure S7A). These results suggest drug passage through the selectivity filter from the external side is less chemically favorable and cannot confer state-dependent inhibition based on its less dynamic and wide aperture ($R = 1.95\text{--}2.6$ Å). Taken together, we propose that uncharged local anesthetics and class 1C antiarrhythmics can access the lateral fenestration receptor site through hydrophobic or

hydrophilic pathways— either by traversing through the membrane fraction or maneuvering into the pore vestibule when the intracellular gate opens.

SUMMARY

Using high throughput electrophysiology, we identified a chemically diverse group of potent PKD2L1 inhibitors, including several with known voltage-gated ion channel pharmacology. In-silico docking analysis and functional mutagenesis identified fenestrations within the PKD2L1 pore domain as a binding site for these compounds. Previous studies of TRP channel pharmacology have highlighted unique ligand binding pockets within their voltage-sensor (or voltage-sensor like) and cytosolic domains, but receptors within the cavities of the pore vestibule have not been previously reported^{50–52}. Polycystins are the earliest-evolved subfamily of TRP channels, with orthologs present in the genomes of bacteria, yeast, and protists. These channels diverged from voltage-gated channels ≈ 750 million years ago⁵³. Based on their common evolutionary origins, it is reasonable to expect that along with several conserved structural features, that polycystins would also share pharmacological sensitivities with Na_v and Ca_v channels⁵⁴. It is important to note that lateral fenestrations and other structural features which confer voltage-dependent gating (e.g., gating charges) found in the polycystin subclass are not present in other subfamilies of TRP channels. Thus, these features are unique to polycystins and can be leveraged to develop selective agents that state-dependently target members of this TRP subclass. Importantly, the PKD2L1 lateral fenestration site (e.g. F549) is also conserved in PKD2— whose germline variants are associated with 15–20% of patients with ADPKD. Notably, we did not identify any polycystin channel activators despite screening a diverse chemical library that included both ion channel activators and inhibitors. This lack of identified activators may be the result of screening compounds at 1 μM concentrations, which sets a stringent threshold in identifying channel activators with high potency. Given that ADPKD is considered a loss of function disease, the therapeutic value for the identified polycystin is uncertain and future screens may benefit from less restrictive potency criteria and testing a larger drug screening library. Nonetheless, our study clearly establishes the utility of the HTS electrophysiology approach for PKD2L1, and the opportunity to identify prototypic polycystic drug activators for research and po.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Abbreviations:

TRP	Transient receptor potential
ADPKD	autosomal dominant polycystic kidney disease

PKD1	Polycystic Kidney Disease 1 gene or protein
PKD1L1	Polycystic Kidney Disease 1 Like 1 gene or protein
PKD1L2	Polycystic Kidney Disease 1 Like 2 gene or protein
PKD2	Polycystic Kidney Disease 2 gene or protein
PKD2L1	Polycystic Kidney Disease 2 Like 1 gene or protein
HTS	high-throughput screen
IC₅₀	half current inhibition
Na_v	voltage-gated sodium channels
Na_v	voltage-gated sodium channel
Ca_v	calcium channel
CDM	calcium-dependent modulation
CDD	irreversible calcium-dependent desensitization
Ki	inhibitory constant
cryo-EM	cryo-electron microscopy
Gd³⁺	gadolinium ion
LA	local anesthetic
AA	class 1C antiarrhythmic
LF	lateral fenestrations
TOP	tetragonal opening for polycystins
SF	selectivity filter

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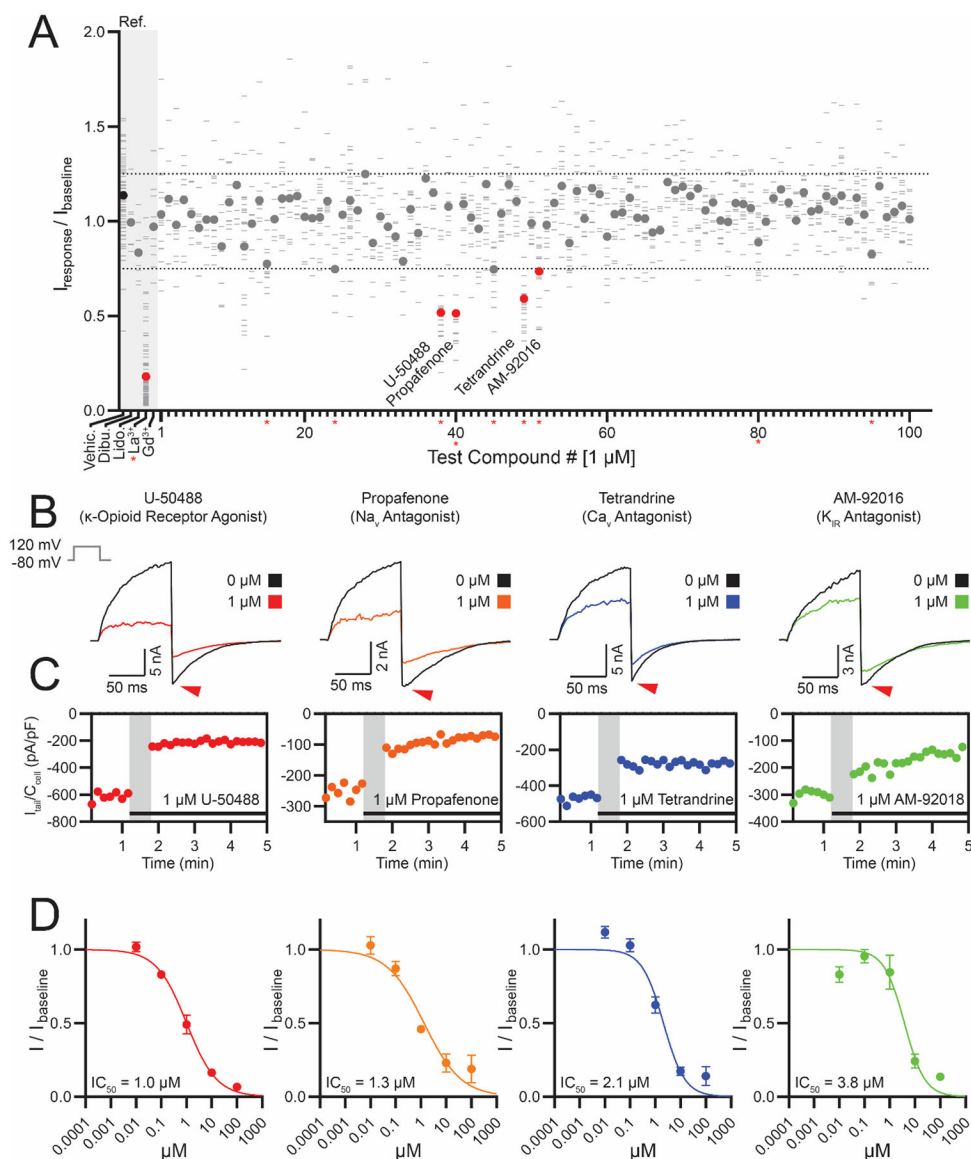


Figure 1. High-throughput drug screen identifies four potent PKD2L1 inhibitors.

(A) Mean post-treatment F514A tail currents recorded through automated-patch clamp, normalized to pre-treatment amplitude. Dashes denote individual cell responses, while circles denote means. Asterisks highlight compounds with a statistically significant change in amplitude relative to currents from vehicle treated cells ($p < 0.0005$, unpaired t-test). Dotted lines denote a $\pm 25\%$ shift from baseline current. Notably, Vehic. response demonstrates residual CDM, showing a 13.6% increase from baseline. Grey overlay distinguishes vehicle control (DMSO) and previously identified PKD2L1 inhibitors (Dibucaine, Lidocaine, La^{3+} n = 7–109 cells) from test compounds (n = 6–38 cells). (B) Exemplar current traces for all four screen-identified hits in APC configuration, before and after treatment with 1 μ M of compound. Inset denotes stimulus protocol used for APC experiments. (C) Exemplar time courses of block by screen-hits. Grey overlay represents liquid handler dosing period, during which stimulus responses were not collected. Black bar

indicates the presence of test compounds in extracellular solution. Due to the likelihood of increased leak and cell loss with subsequent changes in volume, washout steps were not included in APC. **(D)** Drug concentration-response relationships for F514A when treated with screen-identified inhibitors from **A**. Data were fit to the Hill equation to calculate IC_{50} values. (n = 6–19 cells, Error = SEM)

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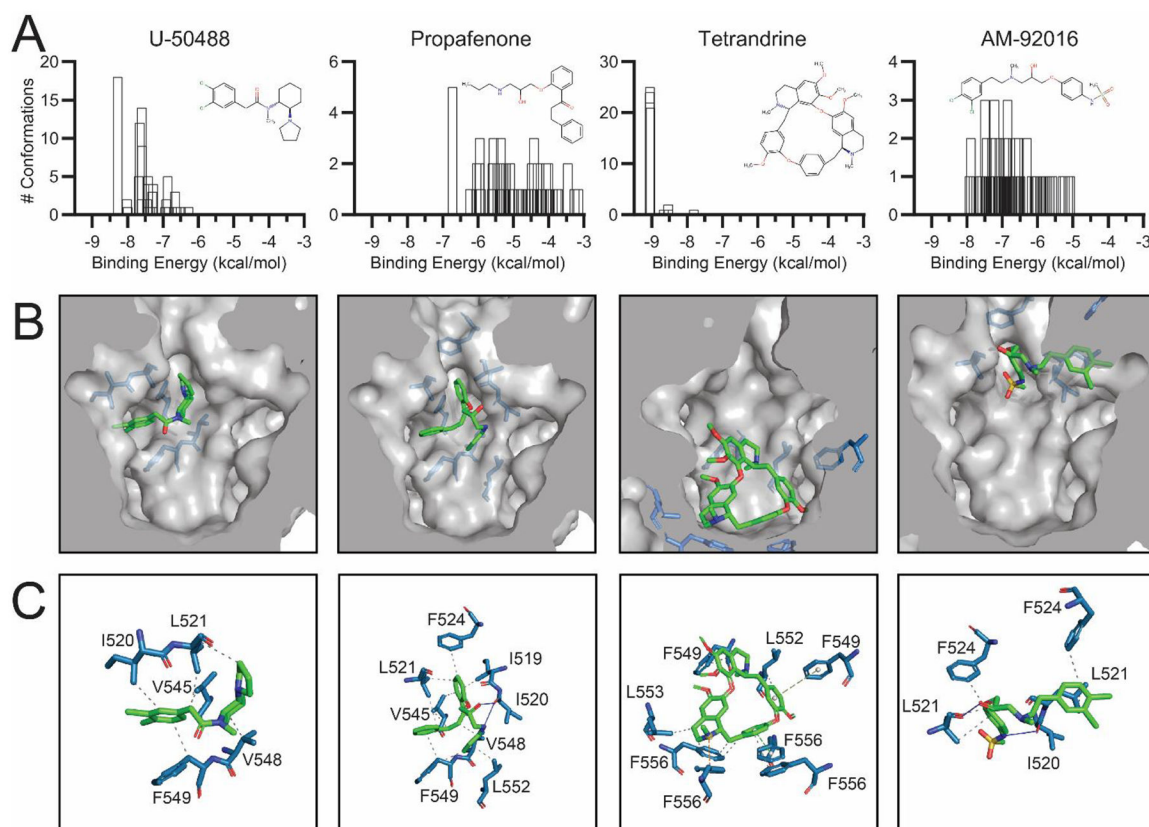


Figure 2. In silico docking predicts drug binding within the PKD2L1 pore vestibule.

(A) Clustering histograms of screen hits. Histograms represent the results from 100 runs of docking analysis against PKD2L1 (PDB: 6DU8), grouped by conformational clusters and ranked according to the lowest binding energy in each cluster. Insets present the chemical structure for each screen hit. (B) Surface representations of highest-ranked docking conformation for each screen hit. (D) Stick representations of molecular interactions between screen hits and binding site residues, analyzed via PLIP.

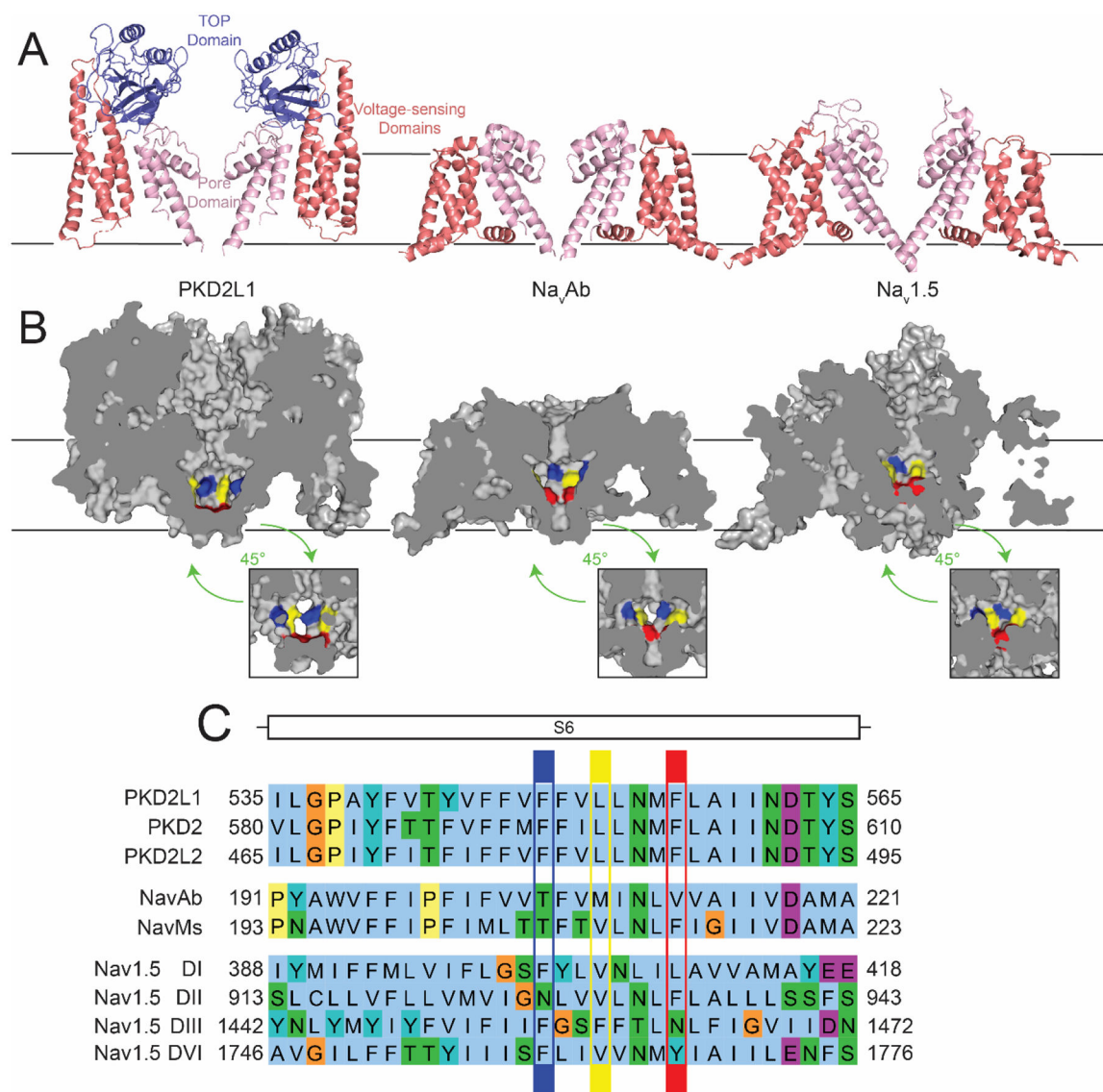


Figure 3. Structural homology between PKD2L1 and Na_v channels implies shared drug binding properties.

(A) Cartoon representations of two isolated domains from PKD2L1, Na_vAb, and Na_v1.5 high-resolution structures, with channel regions colored to highlight the pore domain (pink), voltage-sensing domains (red), and TOP domain (blue)^{7,55,56}. (B) Cross section-view of PKD2L1, Na_vAb, and Na_v1.5 surface representations. Insets provide alternate viewing angle of the pore domain's interior surface. Colors denote locations of Na_vAb drug-binding residues: T206 *blue*, M209 *yellow*, V213 *red*. (C) Sequence alignment of the S6 transmembrane segments of PKD2L1, PKD2, Na_vAb, Na_vMs, and Na_v1.5. Sequence residues colored according to ClustalX scheme, with highlights corresponding to colored residues in B.

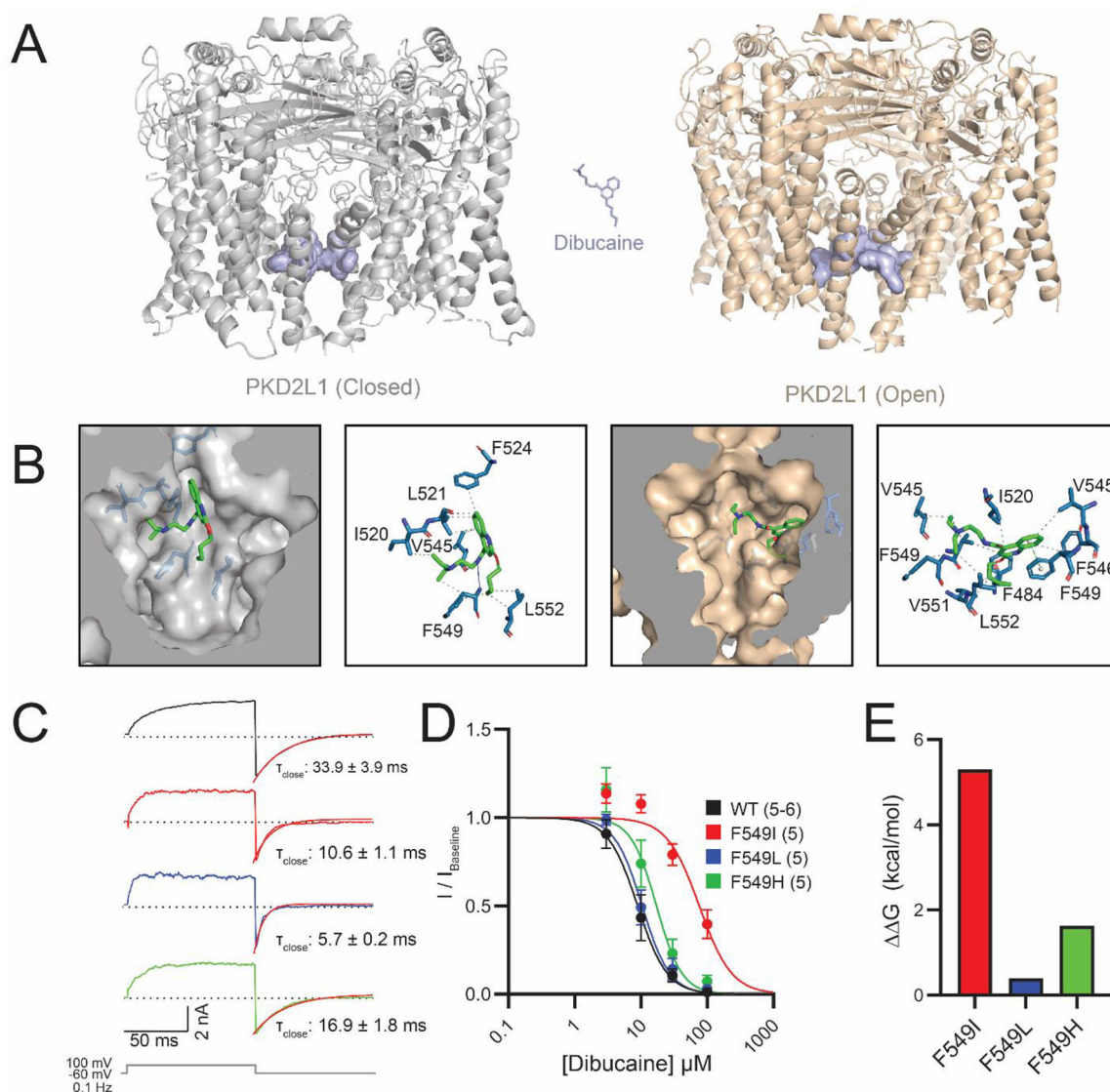


Figure 4. PKD2L1 contains a Na_v-like local dibucaine binding site.

(A) Cartoon representations of PKD2L1 in closed (PDB: 6DU8) and open (PDB: 5Z1W) conformations^{6,7}. Blue surfaces highlight the ten highest ranked docking positions obtained for dibucaine in Autodock4.2. (B) Surface representations of the highest-ranked binding positions for dibucaine. Stick representations of proposed molecular interactions between dibucaine and binding site residues, analyzed with PLIP. (C) Manually recorded sample current traces from WT PKD2L1 and functional F549 mutations, with tail currents fit to a one-phase model of exponential decay. Inset values represent mean closure rates (τ_{close}) for each PKD2L1 variant ($n = 5-6$ cells, Error = SEM) (D) Concentration-response curves for dibucaine inhibition of PKD2L1 WT and F549 mutant currents. ($n = 5-6$ cells, Error = SEM). (E) Change in dibucaine binding affinity for functional F549 mutants, calculated from the change in IC_{50} values relative to WT PKD2L1, with exact values listed in Table S4.

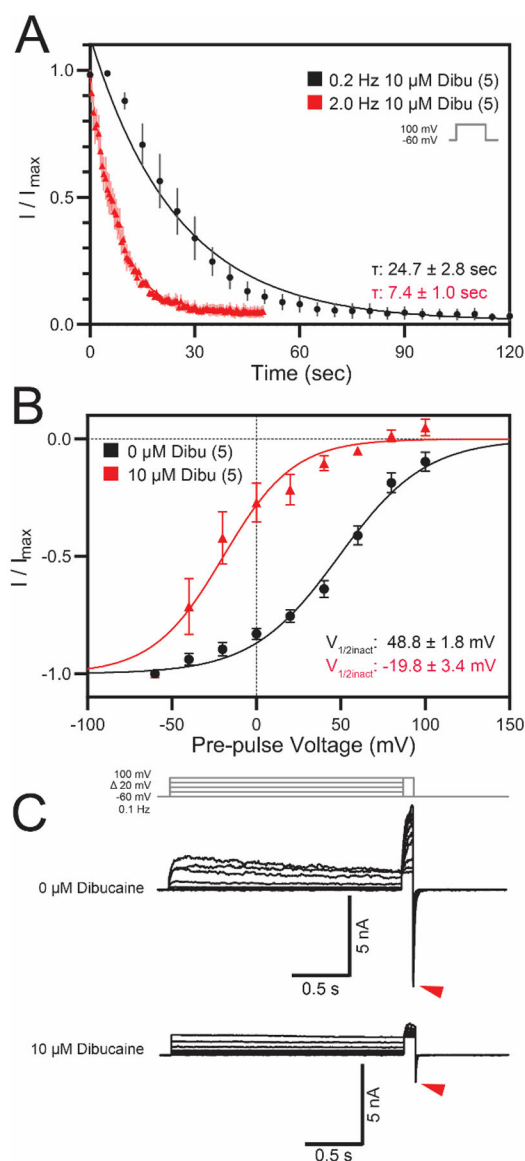


Figure 5. Dibucaine shows use and voltage-dependent inhibition of WT PKD2L1.

(A) Time course of WT PKD2L1 inhibition by 10 μ M dibucaine at 0.2 and 2 Hz sweep frequencies in manual configuration. Time constants (τ) were calculated by fitting tail currents to a one-phase model of exponential decay. Inset denotes stimulus protocol used to generate tail currents. ($n = 5$ cells, Error = SEM) (B) Normalized current-voltage relationship for PKD2L1 when treated externally with control and dibucaine conditions in manual configuration. $V_{1/2inact}$ for each condition was estimated by fitting data to a two-state Boltzmann function. ($n = 5$ cells, Error = SEM) (C) Exemplar current traces from B, showing pre- and post-treatment currents. Inset denotes pre-pulse protocol used to measure $V_{1/2inact}$.

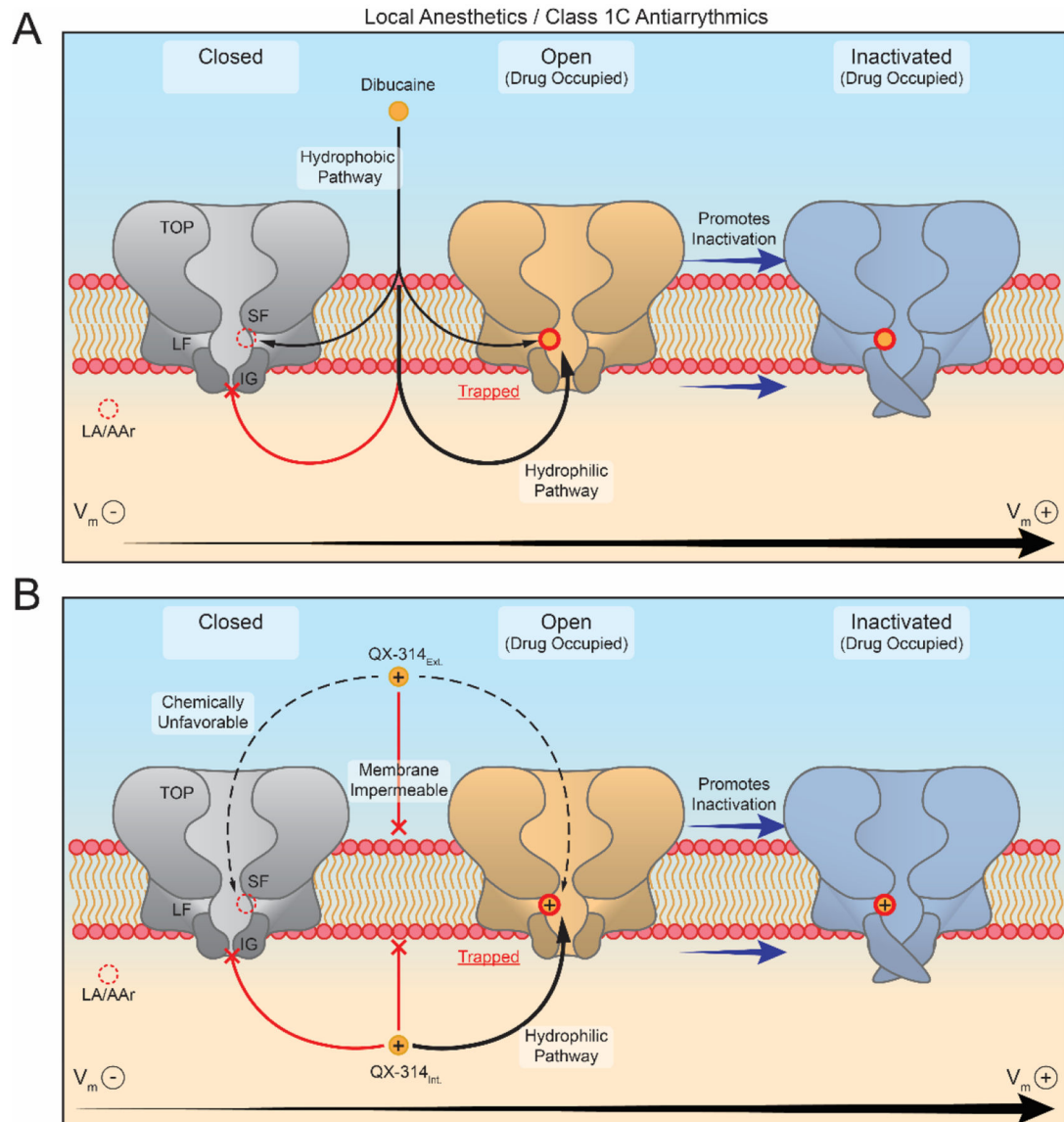


Figure 6. A hypothetical model of drug access to the PKD2L1 lateral fenestration binding site. (A) LA-AA receptor access to the lateral fenestration binding site preferentially occurs via the inner gate (IG), as evidenced by enhanced inhibition of PKD2L1 during high-frequency opening events. For drugs with high lipid solubility, the lateral fenestrations (LF) present an additional hydrophilic access path to the LF receptor, as in Na_v s. (B) Membrane-impermeable local anesthetics (QX-314) show bilateral access to the local anesthetic binding site, with intracellular treatment producing near instantaneous inhibition, while extracellular treatment produces exceedingly slow inhibition. Due to QX-314's inability to traverse through the cell membrane, this extracellular inhibition indicates a path through the tetragonal opening for polycystins (TOP) and selectivity filter (SF) to access the LF receptor. Due to the delayed inhibition produced by extracellular QX-314, this SF pathway is likely much less chemically favorable than the intracellular hydrophilic pathway.